

Transformations of $\sqrt{x}$ - Worksheet

MCR3U

SOLUTIONS

Key points of
 $y = \sqrt{x}$

x	y
0	0
1	1
4	2
9	3

1) 1) State the transformations to the parent function $f(x) = \sqrt{x}$ in the order that you would do them.

a) $f(x) = 2\sqrt{x+1} - 3$

- vertical stretch by 2
- shift left 1 unit
- shift down 3 units

b) $f(x) = 3\sqrt{\frac{1}{2}(x-5)} + 4$

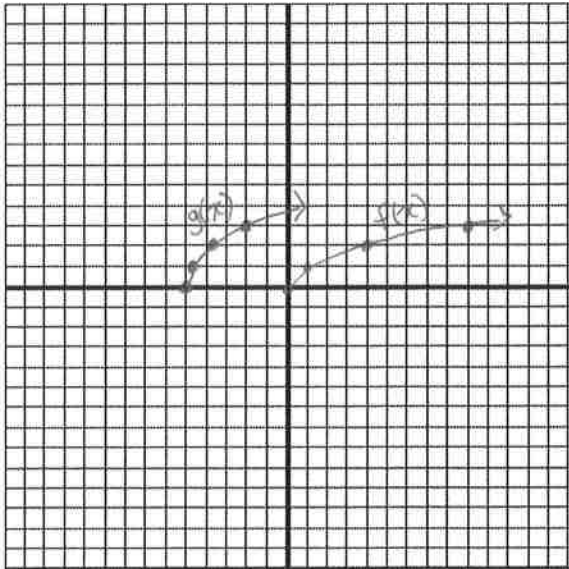
- vertical stretch by 3
- horizontal stretch by 2
- shift right 5 units
- shift up 4 units

c) $f(x) = -\frac{1}{2}\sqrt{-3(x)} - 6$

- vertical compression by $\frac{1}{2}$
- vertical reflection
- horizontal compression by $\frac{1}{3}$
- horizontal reflection
- shift down 6 units.

2) Graph the parent function, $f(x) = \sqrt{x}$. Describe the transformations in order, make a table of values of image points, write the equation of the transformed function and graph it.

a) $g(x) = f[3(x + 5)]$



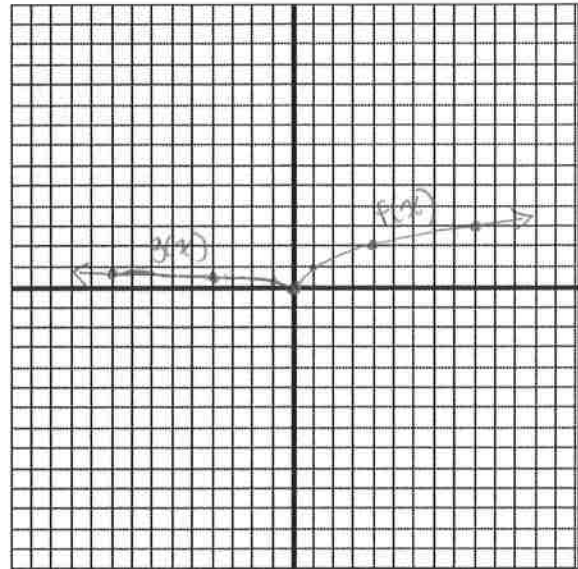
1) horizontal compression by $\frac{1}{3}$ ($\frac{x}{3}$)

2) shift left 5 units ($x - 5$)

$f(x)$	$g(x)$	y
	$\frac{x}{3} - 5$	
(0,0)	-5	0
(1,1)	-4.7	1
(4,2)	-3.7	2
(9,3)	-2	3

$$g(x) = \sqrt{3(x+5)}$$

b) $g(x) = \frac{1}{4}f(-x)$



1) vertical compression by $\frac{1}{4}$ ($\frac{y}{4}$)

2) horizontal reflection ($-x$)

$f(x)$	$g(x)$	y
	$-x$	$\frac{y}{4}$
(0,0)	0	0
(1,1)	-1	0.25
(4,2)	-4	0.5
(9,3)	-9	0.75

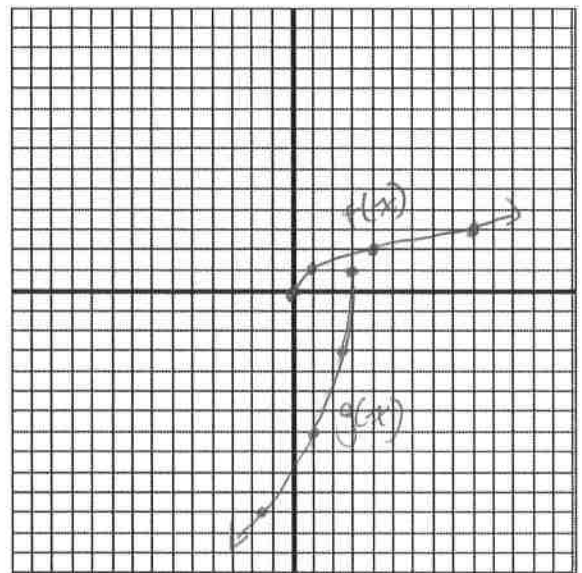
$$g(x) = \frac{1}{4} \sqrt{-x}$$

c) $g(x) = -4f[-2(x-3)] + 1$

- vertical stretch factor 4; vertical reflection $(-4y)$
- horizontal compression factor $\frac{1}{2}$; horizontal reflection $(\frac{x}{-2})$
- shift right 3 units $(x+3)$
- shift up 1 unit $(y+1)$

$f(x)$	$g(x)$
$\frac{x}{2} + 3$	$-4y + 1$
(0,0)	3 1
(1,1)	2.5 -3
(4,2)	1 -7
(9,3)	-1.5 -11

$$g(x) = -4\sqrt{-2(x-3)} + 1$$



3) Use the description to write the transformed function, $g(x)$.

a) The parent function $f(x) = \sqrt{x}$ is compressed vertically by a factor of $\frac{1}{3}$ and then translated (shifted) 3 units left.

$$a = \frac{1}{3}$$

$$d = -3$$

$$g(x) = \frac{1}{3}\sqrt{x+3}$$

b) The parent function $f(x) = \sqrt{x}$ is reflected over the x-axis, stretch horizontally by a factor of 3 and then translated 1 unit left and 4 units down.

$$a = -1$$

$$k = \frac{1}{3}$$

$$d = -1$$

$$c = -4$$

$$g(x) = -1\sqrt{\frac{1}{3}(x+1)} - 4$$

